

Inductive and Deductive Reasoning

Jonathan R. Bacolod

Reasoning is the process of drawing conclusions from facts, observations, or known statements.

Inductive Reasoning

Inductive reasoning begins with specific observations or patterns and forms a general conclusion.

Specific observations $\longrightarrow$ General conclusion

Inductive Conclusions

Conclusions obtained through **inductive reasoning** are reasonable but are not guaranteed to be true.

Deductive Reasoning

Deductive reasoning begins with accepted facts, definitions, postulates, or theorems to arrive at a logically certain conclusion.

Accepted facts $\longrightarrow$ Certain conclusion

Conjecture

A **conjecture** is a statement believed to be true based on observed patterns but has not yet been proven.

Proof

A **proof** is a logical argument that establishes the truth of a mathematical statement using deductive reasoning.

Counterexample

Counterexamples can disprove a conjecture formed through inductive reasoning.

Deductive Proofs

In mathematics, deductive reasoning is used to establish valid proofs.

Examples

Example 1

After observing that

$$2 + 4 = 6, \quad 4 + 6 = 10, \quad 6 + 8 = 14,$$

Maria concludes that the sum of any two consecutive even numbers is divisible by 2.

The conclusion is based on observed examples.

Inductive

Example 2

All rectangles have four right angles.

Figure $ABCD$ is a rectangle.

Therefore, Figure $ABCD$ has four right angles.

The conclusion follows from an accepted fact.

Deductive

Example 3

The first five multiples of 9 have digit sums equal to 9, so every multiple of 9 has a digit sum divisible by 9.

The conclusion is based on observed patterns.

Inductive

Example 4

Vertical angles are congruent.

$\angle 1$ and $\angle 2$ are vertical angles.

Therefore,

$$\angle 1 \cong \angle 2.$$

The conclusion follows from a known theorem.

Deductive

Example 5

The first four terms of the sequence are

3, 6, 9, 12.

Therefore, the next term is 15.

The conclusion is based on an observed pattern.

Inductive

Example 6

A student observes that the interior angle sums of a triangle, quadrilateral, and pentagon are

$$180^\circ, \quad 360^\circ, \quad 540^\circ.$$

The student concludes that the interior angle sum of an n -gon follows a pattern.

Inductive

Example 7

If two lines are parallel, then corresponding angles are congruent.

Lines m and n are parallel.

Therefore, corresponding angles are congruent.

The conclusion logically follows from the given statements.

Valid

Example 8

Every square is a rectangle.

Figure $PQRS$ is a square.

Therefore, Figure $PQRS$ is a rectangle.

The conclusion logically follows from the given statements.

Valid

Practice Exercises

Practice 1

After observing that

$$2 + 4 = 6, \quad 4 + 6 = 10, \quad 6 + 8 = 14,$$

Maria concludes that the sum of any two consecutive even numbers is divisible by 2.

Determine whether the reasoning is **Inductive** or **Deductive**.

Practice 2

All rectangles have four right angles.

Figure $ABCD$ is a rectangle.

Therefore, Figure $ABCD$ has four right angles.

Determine whether the reasoning is **Inductive** or **Deductive**.

Practice 3

The first five multiples of 9 have digit sums equal to 9, so every multiple of 9 has a digit sum divisible by 9.

Identify whether the conclusion is based on **In-**
ductive or **Deductive** reasoning.

Practice 4

Vertical angles are congruent.
 $\angle 1$ and $\angle 2$ are vertical angles.
Therefore,

$$\angle 1 \cong \angle 2.$$

Identify whether the reasoning is **Inductive** or **Deductive**.

Practice 5

The first four terms of the sequence are

3, 6, 9, 12.

Therefore, the next term is 15.

State whether the reasoning is **Inductive** or **Deductive**.

Practice 6

A student observes that the interior angle sums of a triangle, quadrilateral, and pentagon are

$$180^\circ, \quad 360^\circ, \quad 540^\circ.$$

The student concludes that the interior angle sum of an n -gon follows a pattern.

Is this **Inductive** or **Deductive** reasoning?

Practice 7

If two lines are parallel, then corresponding angles are congruent.

Lines m and n are parallel.

Therefore, corresponding angles are congruent.

Determine whether the argument is **valid** or **not valid**.

Practice 8

Every square is a rectangle.

Figure $PQRS$ is a square.

Therefore, Figure $PQRS$ is a rectangle.

Determine whether the argument is **valid** or **not valid**.

Activities

Activity 1

After observing that

$$5 + 7 = 12, \quad 7 + 9 = 16, \quad 9 + 11 = 20,$$

Leo concludes that the sum of any two consecutive odd numbers is divisible by 4.

Determine whether the reasoning is **Inductive** or **Deductive**.

Activity 2

All equilateral triangles have three congruent sides.

Triangle XYZ is equilateral.

Therefore, Triangle XYZ has three congruent sides.

Determine whether the reasoning is **Inductive** or **Deductive**.

Activity 3

The first five square numbers are

odd, even, odd, even, odd.

Therefore, square numbers alternate between odd and even.

Identify whether the reasoning is **Inductive** or **Deductive**.

Activity 4

Supplementary angles have a sum of 180° .

$\angle A$ and $\angle B$ are supplementary.

Therefore,

$$m\angle A + m\angle B = 180^\circ.$$

Identify whether the reasoning is **Inductive** or **Deductive**.

Activity 5

The first four terms of the sequence are

8, 12, 16, 20.

Therefore, the next term is 24.

State whether the reasoning is **Inductive** or **Deductive**.

Activity 6

A student observes that the exterior angle sums of a triangle, quadrilateral, pentagon, and hexagon are all 360° .

The student concludes that the exterior angle sum of every polygon is 360° .

Is this **Inductive** or **Deductive** reasoning?

Activity 7

If two angles are complementary, then their measures add up to 90° .

$\angle P$ and $\angle Q$ are complementary.

Therefore,

$$m\angle P + m\angle Q = 90^\circ.$$

Determine whether the argument is **valid** or **not valid**.

Activity 8

Every rhombus is a parallelogram.

Figure $JKLM$ is a rhombus.

Therefore, Figure $JKLM$ is a parallelogram.

Determine whether the argument is **valid** or **not valid**.

Thank you!